Beaconize this: Genomic data resources with GA4GH and ELIXIR standards

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GA4GH standards and protocols - some co-developed through ELIXIR - have reached deployment grade stability and can drive data resources and networks

- Since 2016 ELIXIR has driven development of the G4GH Beacon protocol
- Beacon v2 was designed with support for rich biomedical data queries and data delivery, in distributed and standalone environments.
- Other mature GA4GH standards with impact on federated genomics are VRS and Phenopackets v2.
- The Progenetix cancer genomics resource tests & demonstrates implementation of these standards in real-world scenarions.

Can you provide data about focal deletions in CDKN2A in Glioblastomas from juvenile patients with unrestricted access?

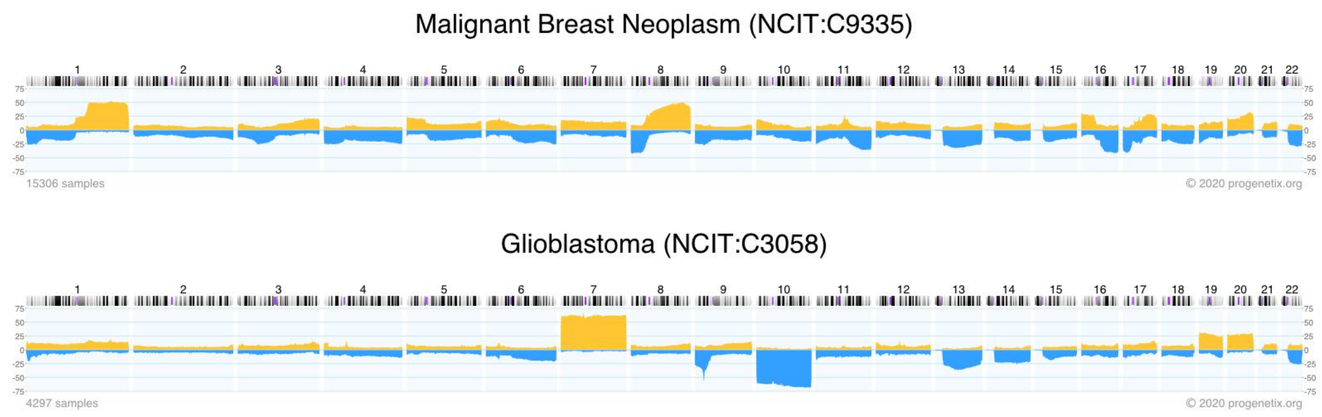
Beacon *v2* API

The Beacon API v2 represents a simple but powerful **genomics API** for ***federated*** data discovery and retrieval

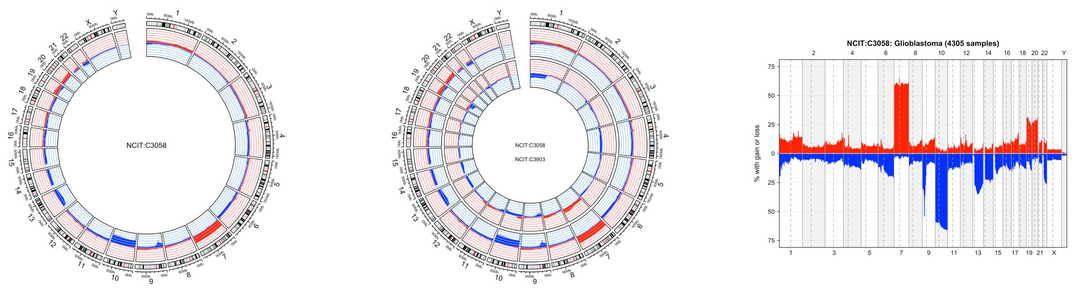
Progenetix and *cancercelllines.org* are oncogenomic resources built using GA4GH standards such as Beacon v2, VRS and Phenopackets v2

- Progenetix is an open oncogenomic reference resource with > 140'000 CNV profiles from cancers and reference samples.
- cancercelllines.org provides annotated genome variants for >16000 cell lines and CNV profiles for >5000 cell line samples
- The sites use the Beacon v2 protocol and data models for communication and empower access to the data in federated scenarios powered by Beacon v2 aggregators
- The *bycon* software stack implements the Beacon v2 with extensions, e.g. by supporting the GA4GH ***Phenopackets*** format as well as various aggregated exports over ***handover*** objects.

Regional CNV Frequencies for Most Cancer Types provided over Beacon *handover* objects



pgxRpi is a, well, Progenetix R API using *handover* objects



Visualization of CNV features using the pgxRpi R package. Aggregated CNV data for cancer typesis displayed using Circos or frequency plots in a local R environment. The R package relies on the Beacon v2 API to communicate with Progenetix.

Links and Info progenetix.org | docs.genomebeacons.org | github.com/progenetix/pgxRpi | github.com/progenetix/bycon